# Assignment 4: SDN

## 1. Update Ubuntu:

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo apt update && sudo apt upgrade -y
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [656 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [844 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [192 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [571 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [62.7 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [119 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [376 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
Fetched 3,590 kB in 4s (821 kB/s)
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  python3-pip-whl python3-pip
Learn more about Ubuntu Pro at https://ubuntu.com/pro
Upgrading:
  base-files  linux-generic-hwe-24.04  linux-headers-generic-hwe-26.04  linux-libc-dev  linux-tools-common
  bpftool     linux-generic-hwe-26.04  linux-image-generic-hwe-26.04    linux-perf

Installing dependencies:
  linux-headers-7.0.0-43          linux-main-modules-v4l2loopback-7.0.0-43-generic  linux-tools-7.0.0-43
  linux-headers-7.0.0-43-generic  linux-main-modules-zfs-7.0.0-43-generic           linux-tools-7.0.0-43-generic
  linux-image-7.0.0-43-generic    linux-modules-7.0.0-43-generic

Suggested packages:
  linux-tools

Summary:
  Upgrading: 9, Installing: 8, Removing: 0, Not Upgrading: 0
  Download size: 221 MB
  Space needed: 375 MB / 41.8 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu6.3 [74.7 kB]
```

## 2. Install Mininet:

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo apt install mininet -y
mininet is already the newest version (2.3.0-1.1build1).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-31          linux-image-7.0.0-31-generic           linux-modules-7.0.0-31-generic  linux-tools-7.0.0-31-generic
  linux-headers-7.0.0-31-generic  linux-main-modules-zfs-7.0.0-31-generic  linux-tools-7.0.0-31
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

## 3. Install Open vSwitch

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo apt install openvswitch-switch -y
openvswitch-switch is already the newest version (3.7.1-2).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-31          linux-image-7.0.0-31-generic           linux-modules-7.0.0-31-generic  linux-tools-7.0.0-31-generic
  linux-headers-7.0.0-31-generic  linux-main-modules-zfs-7.0.0-31-generic  linux-tools-7.0.0-31
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

## 4. Check Open vSwitch:

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo systemctl status openvswitch-switch
● openvswitch-switch.service - Open vSwitch
     Loaded: loaded (/usr/lib/systemd/system/openvswitch-switch.service; enabled; preset: enabled)
     Active: active (exited) since Mon 2026-11-09 03:01:45 UTC; 8min ago
 Invocation: efa95084a98041929b2aea6c1c71f751
   Main PID: 1267 (code=exited, status=0/SUCCESS)
   Mem peak: 1.7M
        CPU: 26ms

Nov 09 03:01:44 Mona systemd[1]: Starting openvswitch-switch.service - Open vSwitch...
Nov 09 03:01:45 Mona systemd[1]: Finished openvswitch-switch.service - Open vSwitch.
```

## 5. Test Mininet:

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo mn
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller

*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 s1
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet> exit
*** Stopping 0 controllers

*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
```

## 6. Install Python and Virtual Environment:

```
vboxuser@Mona:~/CloudSchedulingLab$ sudo apt install curl ca-certificates -y
ca-certificates is already the newest version (20260816ubuntu0.26.04.2).
ca-certificates set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-31          linux-image-7.0.0-31-generic            linux-modules-7.0.0-31-generic  linux-tools-7.0.0-31-generic
  linux-headers-7.0.0-31-generic  linux-main-modules-zfs-7.0.0-31-generic  linux-tools-7.0.0-31
Use 'sudo apt autoremove' to remove them.

Installing:
  curl

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 273 kB
  Space needed: 524 kB / 41.3 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 curl amd64 8.18.0-1ubuntu2.8 [273 kB]
Fetched 273 kB in 2s (143 kB/s)
Selecting previously unselected package curl.
(Reading database… 239911 files and directories currently installed.)
Preparing to unpack …/curl_8.18.0-1ubuntu2.8_amd64.deb…
Unpacking curl (8.18.0-1ubuntu2.8)…
Setting up curl (8.18.0-1ubuntu2.8)…
Processing triggers for man-db (2.13.1-1build1)…
vboxuser@Mona:~/CloudSchedulingLab$ curl -LsSf https://astral.sh/uv/install.sh | sh
downloading uv 0.13.7 x86_64-unknown-linux-gnu
installing to /home/vboxuser/.local/bin
  uv
  uvx
everything's installed!

To add $HOME/.local/bin to your PATH, either restart your shell or run:

    source $HOME/.local/bin/env (sh, bash, zsh)
    source $HOME/.local/bin/env.fish (fish)
vboxuser@Mona:~/CloudSchedulingLab$ source $HOME/.local/bin/env
```

```
vboxuser@Mona:~/CloudSchedulingLab$ uv python install 3.10
Installed Python 3.10.22 in 6.19s
 + cpython-3.10.22-linux-x86_64-gnu (python3.10)
vboxuser@Mona:~/CloudSchedulingLab$ cd ~/CloudSchedulingLab
vboxuser@Mona:~/CloudSchedulingLab$ uv venv --python 3.10 ryu-env
Using CPython 3.10.22
Creating virtual environment at: ryu-env
✔ A virtual environment already exists at `ryu-env`. Do you want to replace it? · yes
Activate with: source ryu-env/bin/activate
vboxuser@Mona:~/CloudSchedulingLab$ source ryu-env/bin/activate
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ python --version
Python 3.10.22
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ python -m pip install pip==24.0
/home/vboxuser/CloudSchedulingLab/ryu-env/bin/python: No module named pip
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ uv pip install --python ryu-env/bin/python pip==24.0
Using Python 3.10.22 environment at: ryu-env
Resolved 1 package in 2.06s
Prepared 1 package in 1.01s
Installed 1 package in 10ms
 + pip==24.0
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ python -m pip --version
pip 24.0 from /home/vboxuser/CloudSchedulingLab/ryu-env/lib/python3.10/site-packages/pip (python 3.10)
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ python -m pip install setuptools==67.6.1 wheel
Collecting setuptools==67.6.1
  Using cached setuptools-67.6.1-py3-none-any.whl.metadata (6.2 kB)
Collecting wheel
  Using cached wheel-0.48.0-py3-none-any.whl.metadata (2.3 kB)
Collecting packaging>=24.0 (from wheel)
  Using cached packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Using cached setuptools-67.6.1-py3-none-any.whl (1.1 MB)
Using cached wheel-0.48.0-py3-none-any.whl (33 kB)
Using cached packaging-26.3-py3-none-any.whl (129 kB)
Installing collected packages: setuptools, packaging, wheel
Successfully installed packaging-26.3 setuptools-67.6.1 wheel-0.48.0
```

```
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ python -m pip install --no-build-isolation ryu
Collecting ryu
  Using cached ryu-4.34.tar.gz (1.1 MB)
  Preparing metadata (setup.py) ... done
Collecting eventlet!=0.18.3,!=0.20.1,!=0.21.0,!=0.23.0,>=0.18.2 (from ryu)
  Downloading eventlet-0.41.3-py3-none-any.whl.metadata (5.4 kB)
Collecting msgpack>=0.3.0 (from ryu)
  Downloading msgpack-1.2.4-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (8.3 kB)
Collecting netaddr (from ryu)
  Downloading netaddr-1.3.0-py3-none-any.whl.metadata (5.0 kB)
Collecting oslo.config>=2.5.0 (from ryu)
  Downloading oslo_config-10.4.0-py3-none-any.whl.metadata (2.3 kB)
Collecting ovs>=2.6.0 (from ryu)
  Downloading ovs-4.0.0.tar.gz (155 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 155.1/155.1 kB 1.2 MB/s eta 0:00:00
  Preparing metadata (pyproject.toml) ... done
Collecting routes (from ryu)
  Downloading Routes-2.5.1-py2.py3-none-any.whl.metadata (25 kB)
Collecting six>=1.4.0 (from ryu)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting tinyrpc (from ryu)
  Downloading tinyrpc-1.1.7-py3-none-any.whl.metadata (6.7 kB)
Collecting webob>=1.2 (from ryu)
  Downloading webob-1.8.11-py2.py3-none-any.whl.metadata (13 kB)
Collecting dnspython>=1.15.0 (from eventlet!=0.18.3,!=0.20.1,!=0.21.0,!=0.23.0,>=0.18.2->ryu)
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Collecting greenlet>=1.0 (from eventlet!=0.18.3,!=0.20.1,!=0.21.0,!=0.23.0,>=0.18.2->ryu)
  Downloading greenlet-3.5.6-cp310-cp310-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (3.8 kB)
Collecting stevedore>=5.6.0 (from oslo.config>=2.5.0->ryu)
  Downloading stevedore-5.8.0-py3-none-any.whl.metadata (2.3 kB)
Collecting oslo.i18n>=3.15.3 (from oslo.config>=2.5.0->ryu)
  Downloading oslo_i18n-6.8.0-py3-none-any.whl.metadata (2.0 kB)
Collecting rfc3986>=1.2.0 (from oslo.config>=2.5.0->ryu)
  Downloading rfc3986-2.0.0-py2.py3-none-any.whl.metadata (6.6 kB)
```

```
t/wsgi.py)
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ nano ryu-env/lib/python3.10/site-packages/ryu/app/wsgi.py
```

```
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ python -m pip install --force-reinstall --no-deps ryu==4.34
Collecting ryu==4.34
  Using cached ryu-4.34-py3-none-any.whl
Installing collected packages: ryu
  Attempting uninstall: ryu
    Found existing installation: ryu 4.34
    Uninstalling ryu-4.34:
      Successfully uninstalled ryu-4.34
Successfully installed ryu-4.34
```

```
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ python -c 'from pathlib import Path; p=Path("ryu-env/lib/python3.10/site-packages/ryu/app/wsgi.py"); s=p
.read_text(); old="from eventlet.wsgi import ALREADY_HANDLED"; assert old in s, "Original import not found"; p.write_text(s.replace(old, "ALREADY_HAND
LED = object()", 1)); print("Ryu compatibility fix applied successfully")'
Ryu compatibility fix applied successfully
```

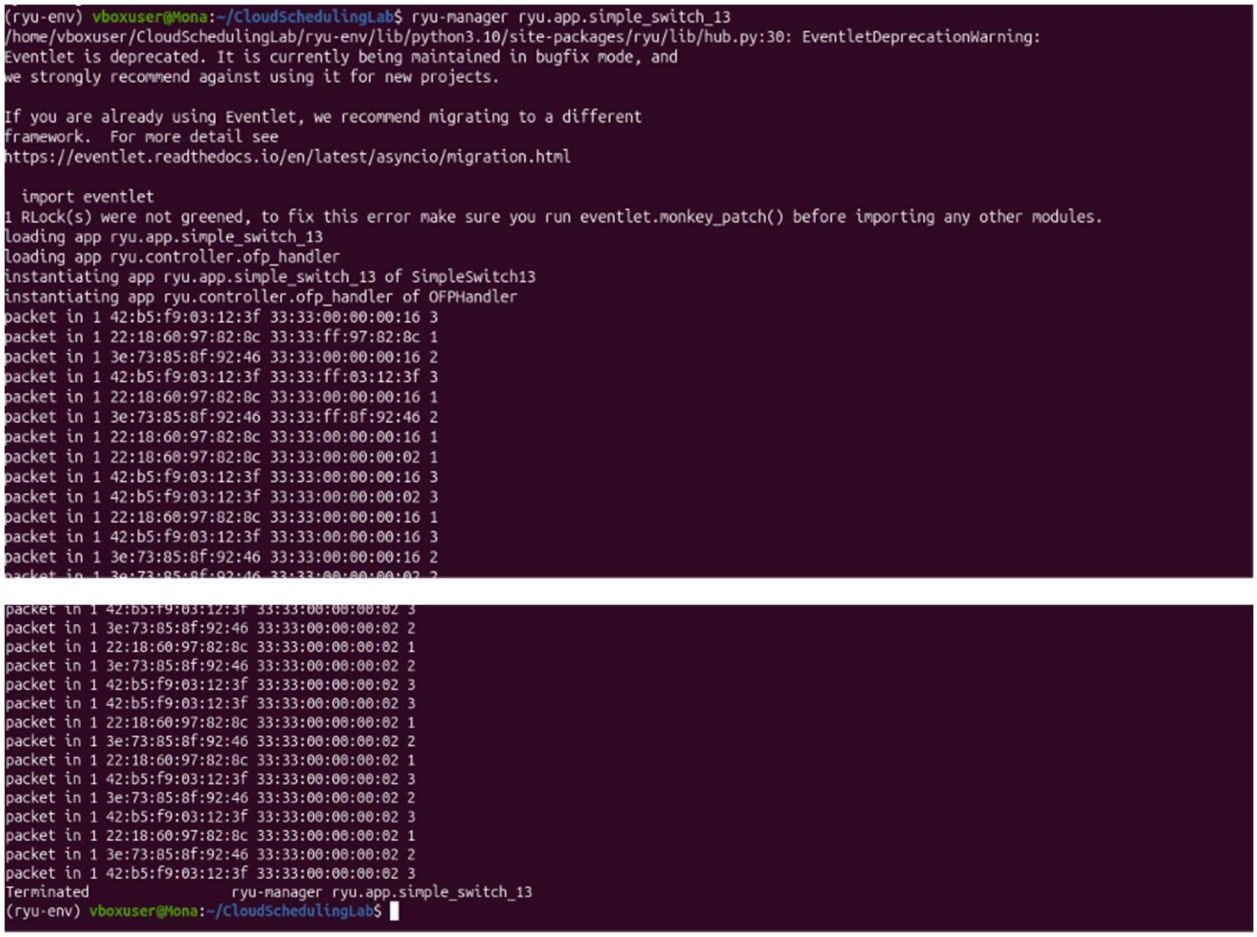
7. Test Ryu Installation:

```
(ryu-env) vboxuser@Mona:~/CloudSchedulingLab$ ryu-manager --version
/home/vboxuser/CloudSchedulingLab/ryu-env/lib/python3.10/site-packages/ryu/lib/hub.py:30: EventletDeprecationWarning:
Eventlet is deprecated. It is currently being maintained in bugfix mode, and
we strongly recommend against using it for new projects.

If you are already using Eventlet, we recommend migrating to a different
framework.  For more detail see
https://eventlet.readthedocs.io/en/latest/asyncio/migration.html

  import eventlet
ryu-manager 4.34
```

8. Start Ryu Controller:







9. Create Mininet SDN Topology:



## 10. Check Nodes:

```
mininet> nodes
available nodes are:
c0 h1 h2 h3 s1
```

## 11. Check Connections:

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
s1 lo:  s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0
c0
```

## 12. Test Connectivity:

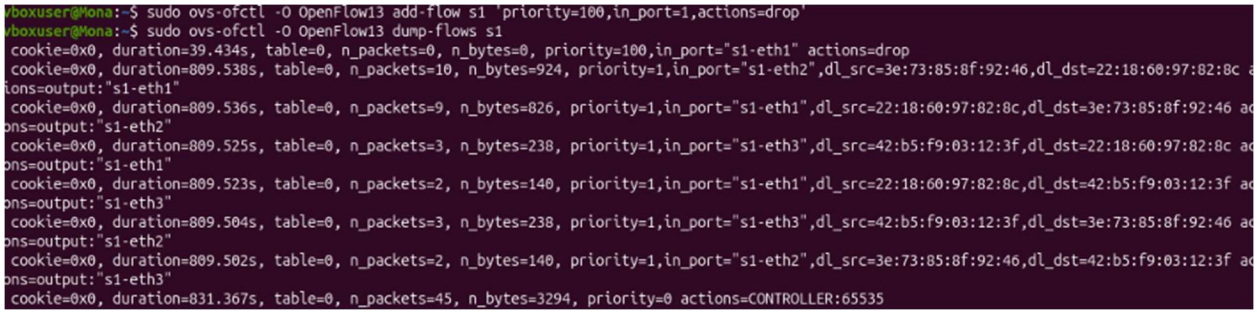
```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.351 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.143 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.095 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.087 ms
^C
--- 10.0.0.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6979ms
rtt min/avg/max/mdev = 0.052/0.128/0.351/0.094 ms
```

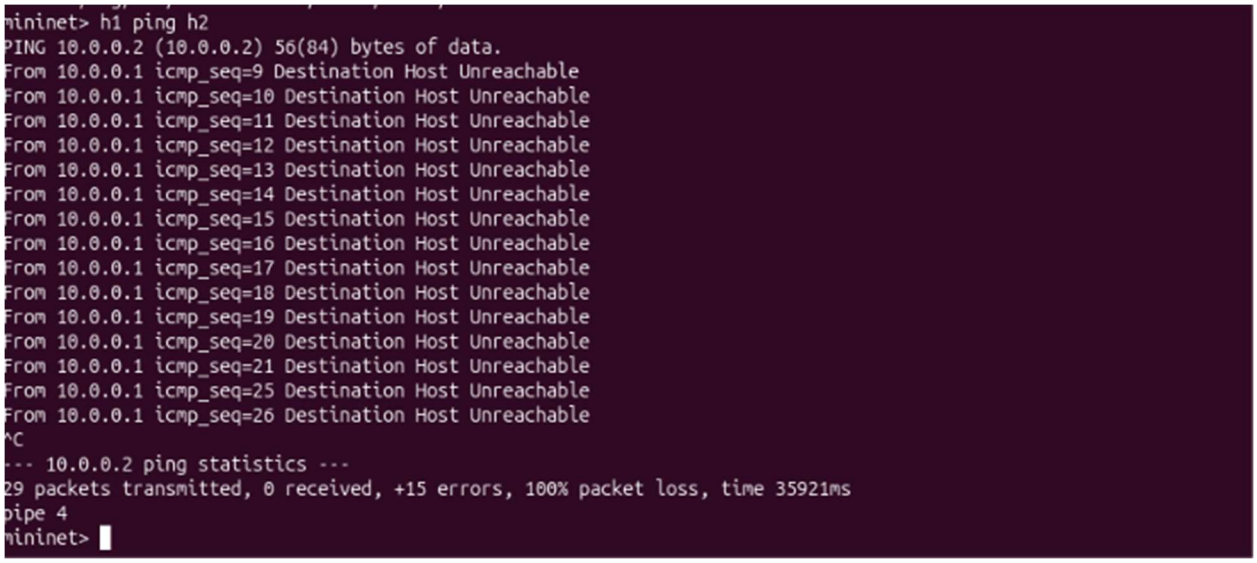
## 13. View Flow Rules:

```
vboxuser@Mona:~$ sudo ovs-ofctl -O OpenFlow13 show s1
[sudo: authenticate] Password:
OFPT_FEATURES_REPLY (OF1.3) (xid=0x2): dpid:0000000000000001
n_tables:254, n_buffers:0
capabilities: FLOW_STATS TABLE_STATS PORT_STATS GROUP_STATS QUEUE_STATS
OFPST_PORT_DESC reply (OF1.3) (xid=0x3):
 1(s1-eth1): addr:06:e2:a6:1a:04:6b
     config:     0
     state:      LIVE
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 2(s1-eth2): addr:8a:34:57:14:d3:15
     config:     0
     state:      LIVE
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 3(s1-eth3): addr:b6:f5:59:37:25:33
     config:     0
     state:      LIVE
     current:    10GB-FD COPPER
     speed: 10000 Mbps now, 0 Mbps max
 LOCAL(s1): addr:16:71:0d:76:37:43
     config:     PORT_DOWN
     state:      LINK_DOWN
     speed: 0 Mbps now, 0 Mbps max
OFPT_GET_CONFIG_REPLY (OF1.3) (xid=0x9): frags=normal miss_send_len=0
```

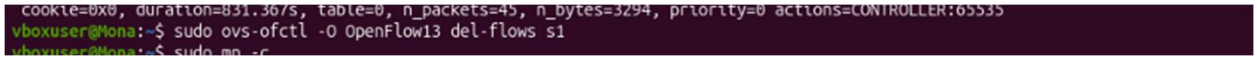
14. Add a Simple Filtering Rule:




15. Test the Filtering Rule:




16. Remove Flow Rules:



17. Clean Mininet:

